

TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING (IOE)

DIGITAL LOGIC

Course Code: ENEX 152 / EX 152 / CT 152

BE Computer (BCT) & BE Electronics (BEI) Engineering

Master Past Examination Solutions Book

Complete Solved Papers with Complete TikZ Logic Schematics, K-Maps & FSM State Automata

Covering All 4 Exam Sessions:

2083 Baishakh, 2082 Bhadra, 2082 Baishakh, and 2081 Ashwin

Author: Antigravity Academic Publishing & Typesetting

Institute of Engineering Central Reference Portal

Contents

1	2083 Baishakh Examination Solutions	3
1.0.1	Question 1: Gray Code to Binary Conversion [2 + 2 = 4 Marks]	3
1.0.2	Question 2: Analog vs. Digital Signals & Excess-3 Code [1 + 2 = 3 Marks]	4
1.0.3	Question 3: Universal Logic Gates Schematics [3 Marks]	4
1.0.4	Question 4: 4-Variable K-Map Minimization & Circuit [5 Marks]	5
1.0.5	Question 5: Octal-to-Binary Encoder Logic Circuit [2 + 4 = 6 Marks]	6
1.0.6	Question 6: Full Subtractor Using Multiplexers [4 Marks]	7
1.0.7	Question 7: D Flip-Flop to JK Flip-Flop Conversion Schematic [2 + 5 = 7 Marks]	8
1.0.8	Question 8: 4-Bit SIPO Shift Register Schematic Timing Diagram [5 Marks]	8
1.0.9	Question 9: Asynchronous BCD (Decade) Counter Schematic [5 Marks]	9
1.0.10	Question 10: Synchronous Mod-10 UP Counter Schematic [7 Marks]	9
1.0.11	Question 11: Synchronous FSM Sequence Detector '011' [7 Marks]	10
1.0.12	Question 12: Characteristics of Digital Logic Families [4 Marks]	10
2	2082 Bhadra Examination Solutions	11
2.0.1	Question 1: Binary to Gray Code Conversion [2 + 2 = 4 Marks]	11
2.0.2	Question 2: 2's Complement & BCD Addition [1 + 2 = 3 Marks]	11
2.0.3	Question 3: De Morgan's Gate Equivalences [4 Marks]	11
2.0.4	Question 4: K-Map Minimization with Don't Cares Circuit [5 Marks]	12
2.0.5	Question 5: 3-Bit Magnitude Comparator Schematic [1 + 5 = 6 Marks]	12
2.0.6	Question 6: Full Adder Using Multiplexers [4 Marks]	13
2.0.7	Question 7: SR to JK Flip-Flop Conversion Schematic [2 + 5 = 7 Marks]	13
2.0.8	Question 8: 4-Bit Johnson Counter Schematic [2 + 3 = 5 Marks]	14
2.0.9	Question 9: 3-Bit Ripple UP Counter with Positive Edge Triggering [5 Marks]	14
2.0.10	Question 10: Synchronous Mod-6 Counter Using SR Flip-Flops [7 Marks]	15
2.0.11	Question 11: Synchronous FSM Sequence Detector '110' [7 Marks]	15
2.0.12	Question 12: Frequency Counter Block Diagram [3 Marks]	16
3	2082 Baishakh Examination Solutions	17
3.0.1	Question 1: ASCII Code & 2's Complement Addition [1 + 3 = 4 Marks]	17
3.0.2	Question 2: Boolean Algebra Proofs [2 + 2 = 4 Marks]	17
3.0.3	Question 3: Octal Priority Encoder Schematic [2 + 6 = 8 Marks]	17
3.0.4	Question 4: 3x8 Decoder Function Realization [3 + 3 = 6 Marks]	18
3.0.5	Question 5: Synchronous 2-Bit UP/DOWN Counter Schematic [7 Marks]	18
3.0.6	Question 6: 4-Bit PISO Shift Register Schematic [3 + 2 = 5 Marks]	19
3.0.7	Question 7: Mod-12 Asynchronous Counter Schematic [3 + 3 = 6 Marks]	19
3.0.8	Question 8: FSM Sequence Detector '010' Using D Flip-Flops [10 Marks]	20
3.0.9	Question 9: Two-Input CMOS NAND Gate Transistor Circuit [4 + 2 = 6 Marks]	21
3.0.10	Question 10: Time Interval Measurement Block Diagram [4 Marks]	22

4	2081 Ashwin Examination Solutions	23
4.0.1	Question 1: Merits & Demerits of Digital Signals [2 Marks]	23
4.0.2	Question 2: Radix Gray Code Conversions [3 Marks]	23
4.0.3	Question 3: BCD to 7-Segment 'f' Segment Decoder Logic Circuit [4 Marks]	23
4.0.4	Question 4: 4:1 MUX Function Realization Combined Adder/Subtractor [3 + 7 = 10 Marks]	24
4.0.5	Question 5: Mod-6 Synchronous DOWN Counter Schematic [7 Marks]	24
4.0.6	Question 6: 4-Bit SIPO Shift Register Schematic [3 + 3 = 6 Marks]	25
4.0.7	Question 7: 3-Bit Asynchronous UP/DOWN Counter Schematic [4 + 3 = 7 Marks]	25
4.0.8	Question 8: FSM Sequence Detector '101' Using SR Flip-Flops [10 Marks]	26
4.0.9	Question 9: Two-Input TTL NOR Gate Transistor Circuit [5 + 2 = 7 Marks]	26
4.0.10	Question 10: Digital Frequency Measurement Block Diagram [4 Marks]	27

Chapter 1

2083 Baishakh Examination Solutions

Examination Overview

Level: Bachelor in Engineering (BE)
Year/Part: I / II

Program: BCT, BEI
Full Marks: 60 **Pass Marks:** 24

1.0.1 Question 1: Gray Code to Binary Conversion [2 + 2 = 4 Marks]

Problem Statement (2083 Baishakh, Q1)

Convert the following Gray codes to binary codes:

- (a) 10011_{Gray}
- (b) 110001_{Gray}

Conversion Algorithm: Let Gray code bits be $G = g_{n-1}g_{n-2} \dots g_0$ and Binary bits be $B = b_{n-1}b_{n-2} \dots b_0$.

- Most Significant Bit (MSB): $b_{n-1} = g_{n-1}$.
- For all subsequent bits ($i = n - 2$ down to 0): $b_i = b_{i+1} \oplus g_i$.

Part (a): 10011_{Gray}

1. $b_4 = g_4 = 1$
2. $b_3 = b_4 \oplus g_3 = 1 \oplus 0 = 1$
3. $b_2 = b_3 \oplus g_2 = 1 \oplus 0 = 1$
4. $b_1 = b_2 \oplus g_1 = 1 \oplus 1 = 0$
5. $b_0 = b_1 \oplus g_0 = 0 \oplus 1 = 1$

Therefore: $(10011)_{\text{Gray}} = (11101)_2$.

Part (b): 110001_{Gray}

1. $b_5 = g_5 = 1$
2. $b_4 = b_5 \oplus g_4 = 1 \oplus 1 = 0$
3. $b_3 = b_4 \oplus g_3 = 0 \oplus 0 = 0$

$$4. b_2 = b_3 \oplus g_2 = 0 \oplus 0 = 0$$

$$5. b_1 = b_2 \oplus g_1 = 0 \oplus 0 = 0$$

$$6. b_0 = b_1 \oplus g_0 = 0 \oplus 1 = 1$$

Therefore: $(110001)_{\text{Gray}} = (100001)_2$.

Verified Final Result

$$\begin{aligned} \text{(a)} \quad (10011)_{\text{Gray}} &= (\mathbf{11101})_2 \\ \text{(b)} \quad (110001)_{\text{Gray}} &= (\mathbf{100001})_2 \end{aligned}$$

1.0.2 Question 2: Analog vs. Digital Signals & Excess-3 Code [1 + 2 = 3 Marks]

Problem Statement (2083 Baishakh, Q2)

Differentiate between analog and digital signals. Convert $(159)_{10}$ to excess-3 code.

Comparison: Analog vs. Digital Signals

Parameter	Analog Signal	Digital Signal
Nature	Continuous in time and amplitude	Discrete in time and quantized in amplitude
Values	Infinite values within a given range	Discrete levels (typically binary '0' and '1')
Noise Immunity	High susceptibility to noise/distortion	Superior noise immunity and error correction
Storage/Processing	Complex analog filters and storage	Easy storage in semiconductor memory, digital D
Example	Natural speech, temperature voltage	Microprocessor signals, digital audio/video

Conversion of $(159)_{10}$ to Excess-3 Code: Excess-3 is formed by adding 3 (0011_2) to each decimal digit:

- Digit 1: $1 + 3 = 4 \Rightarrow 0100_2$
- Digit 5: $5 + 3 = 8 \Rightarrow 1000_2$
- Digit 9: $9 + 3 = 12 \Rightarrow 1100_2$

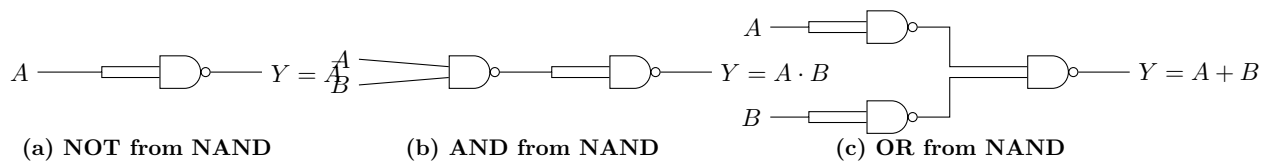
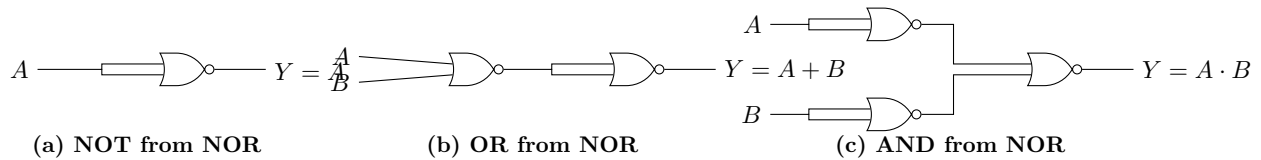
Verified Final Result

$$(159)_{10} = \mathbf{0100 \ 1000 \ 1100}_{\text{Excess-3}}$$

1.0.3 Question 3: Universal Logic Gates Schematics [3 Marks]

Problem Statement (2083 Baishakh, Q3)

Show that NAND and NOR gates are universal gates with logic diagrams.

Logic Schematic: Basic Gates Realized Using NAND Only**Logic Schematic: Basic Gates Realized Using NOR Only****Verified Final Result**

Both NAND and NOR independently synthesize NOT, AND, and OR operators, proving universality.

1.0.4 Question 4: 4-Variable K-Map Minimization & Circuit [5 Marks]**Problem Statement (2083 Baishakh, Q4)**

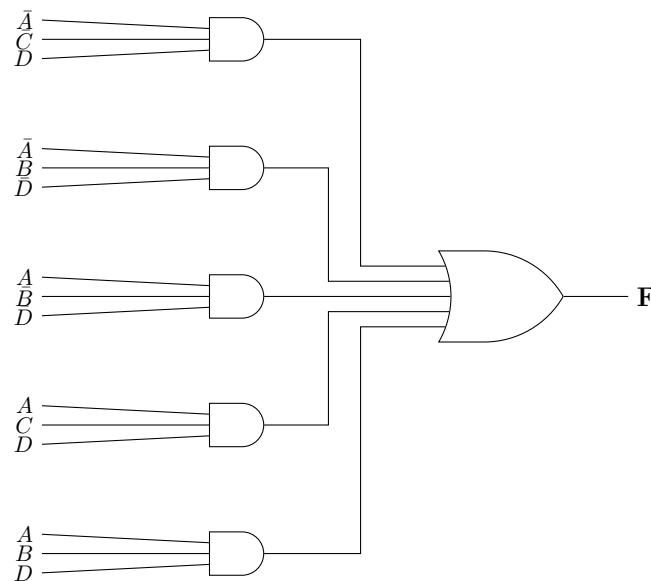
Minimize $F(A, B, C, D) = \sum m(1, 2, 4, 5, 6, 8, 10, 11, 13, 15)$ using K-Map and realize it with suitable logic gates.

Karnaugh Map Plotting:

$AB \backslash CD$	00	01	11	10
00	0 (m_0)	1 (m_1)	0 (m_3)	1 (m_2)
01	1 (m_4)	1 (m_5)	0 (m_7)	1 (m_6)
11	0 (m_{12})	1 (m_{13})	1 (m_{15})	0 (m_{14})
10	1 (m_8)	0 (m_9)	1 (m_{11})	1 (m_{10})

Simplified Boolean Expression:

$$F(A, B, C, D) = \bar{A}\bar{C}D + \bar{A}B\bar{D} + A\bar{B}\bar{D} + ACD + ABD$$

Logic Schematic: Logic Circuit Realization for $F(A, B, C, D)$ **Verified Final Result**

$$F = \bar{A}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{D} + A\bar{B}\bar{D} + A\bar{C}\bar{D} + A\bar{B}D$$

1.0.5 Question 5: Octal-to-Binary Encoder Logic Circuit [2 + 4 = 6 Marks]**Problem Statement (2083 Baishakh, Q5)**

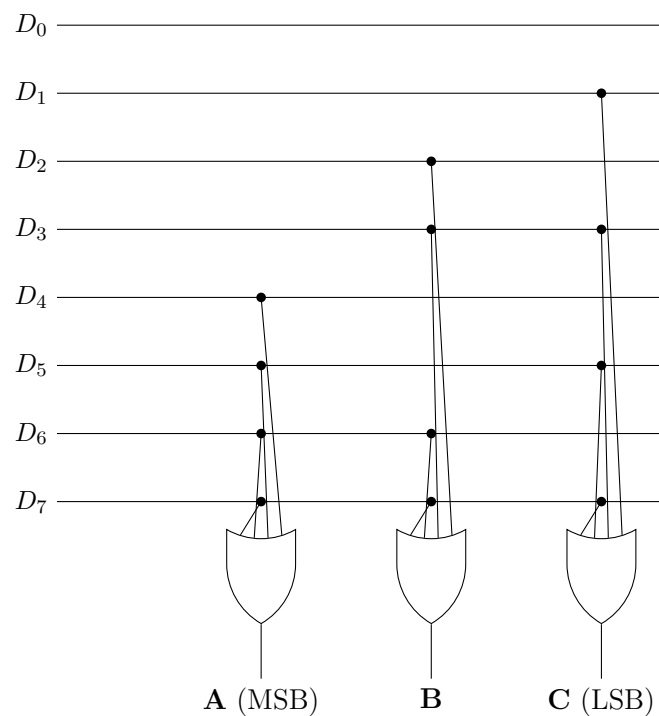
Differentiate between encoder and decoder. Design an Octal-to-Binary Encoder with necessary diagrams.

Boolean Output Equations:

$$A = D_4 + D_5 + D_6 + D_7$$

$$B = D_2 + D_3 + D_6 + D_7$$

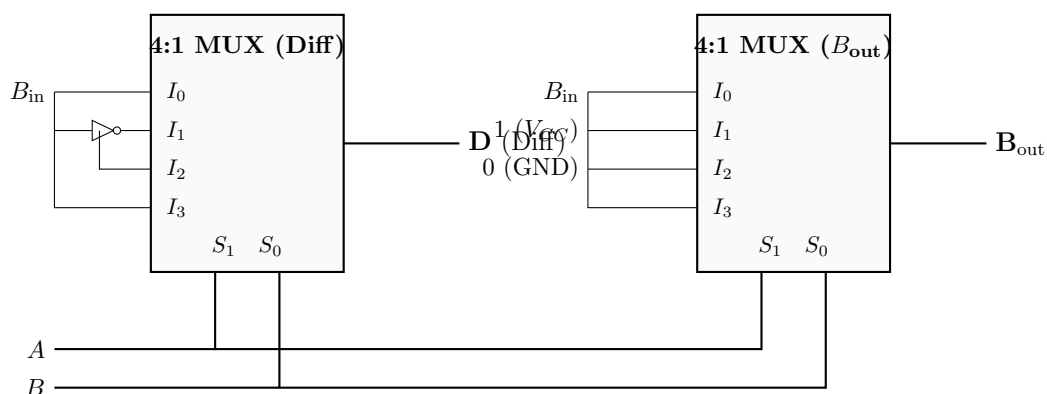
$$C = D_1 + D_3 + D_5 + D_7$$

Logic Schematic: Octal-to-Binary (8-to-3) Encoder Logic Schematic**Verified Final Result**

Logic Equations: $A = D_4 + D_5 + D_6 + D_7$, $B = D_2 + D_3 + D_6 + D_7$, $C = D_1 + D_3 + D_5 + D_7$.

1.0.6 Question 6: Full Subtractor Using Multiplexers [4 Marks]**Problem Statement (2083 Baishakh, Q6)**

Implement a Full Subtractor circuit using Multiplexer.

Logic Schematic: Full Subtractor Realization Using Two 4:1 Multiplexers**Verified Final Result**

Difference (D): $I_0 = B_{in}$, $I_1 = \overline{B_{in}}$, $I_2 = 0$, $I_3 = B_{in}$.

Borrow (B_{out}): $I_0 = B_{in}$, $I_1 = 1$, $I_2 = 0$, $I_3 = B_{in}$. Select lines: $S_1 = A$, $S_0 = B$.

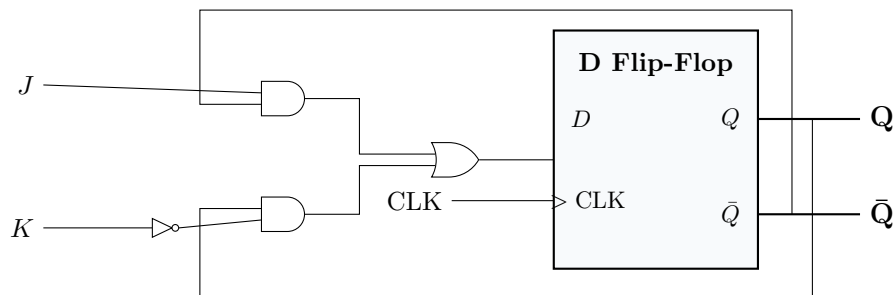
1.0.7 Question 7: D Flip-Flop to JK Flip-Flop Conversion Schematic [2 + 5 = 7 Marks]

Problem Statement (2083 Baishakh, Q7)

Differentiate between latch and flip-flop. Modify a D flip-flop such that it functions as a JK flip-flop.

Conversion Equation: $D = J\bar{Q} + \bar{K}Q$.

Logic Schematic: JK Flip-Flop Implementation Using D Flip-Flop



Verified Final Result

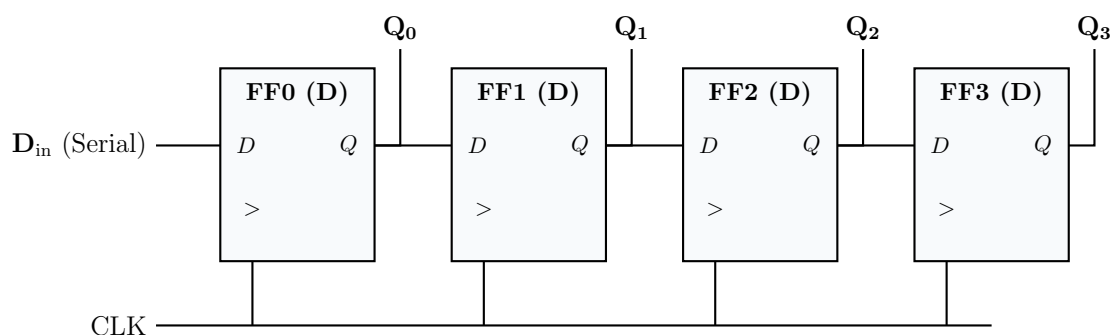
$$D = J\bar{Q} + \bar{K}Q$$

1.0.8 Question 8: 4-Bit SIPO Shift Register Schematic Timing Diagram [5 Marks]

Problem Statement (2083 Baishakh, Q8)

Explain the working of 4-bit SIPO register with timing diagram of 1010 data input.

Logic Schematic: 4-Bit Serial-In Parallel-Out (SIPO) Shift Register Schematic



Verified Final Result

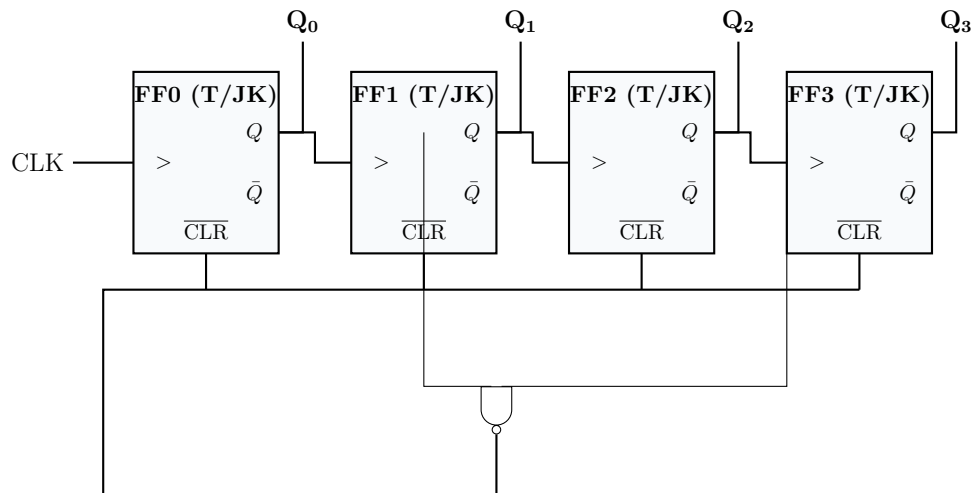
Four clock pulses load serial word 1010_2 into parallel outputs $Q_3Q_2Q_1Q_0$.

1.0.9 Question 9: Asynchronous BCD (Decade) Counter Schematic [5 Marks]

Problem Statement (2083 Baishakh, Q9)

Describe the operation of asynchronous BCD (decade) counter with necessary diagrams.

Logic Schematic: Asynchronous BCD (Decade) Counter Schematic



Verified Final Result

Reset Equation: $\overline{\text{CLR}} = \overline{Q_3} \cdot \overline{Q_1}$. The counter resets at binary state 1010_2 , providing 10 stable states (0 to 9).

1.0.10 Question 10: Synchronous Mod-10 UP Counter Schematic [7 Marks]

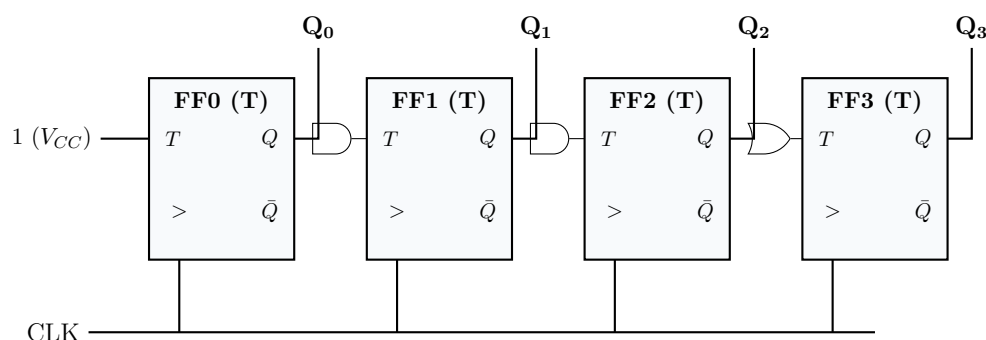
Problem Statement (2083 Baishakh, Q10)

Design a synchronous Mod-10 UP counter using T flip-flops and draw its timing diagram.

Design Equations:

$$T_0 = 1, \quad T_1 = \overline{Q_3}Q_0, \quad T_2 = Q_1Q_0, \quad T_3 = Q_2Q_1Q_0 + Q_3Q_0$$

Logic Schematic: Synchronous Mod-10 UP Counter Logic Schematic

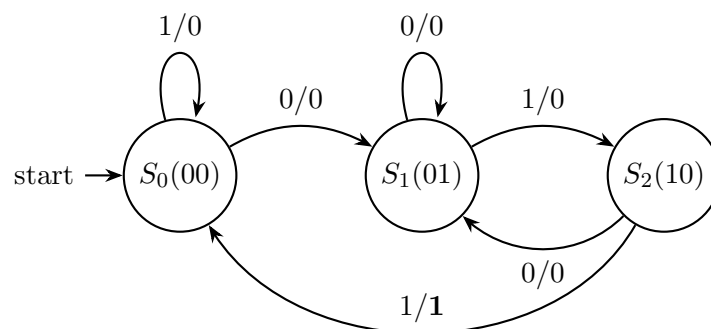


Verified Final Result

$$T_0 = 1, \quad T_1 = \bar{Q}_3 Q_0, \quad T_2 = Q_1 Q_0, \quad T_3 = Q_2 Q_1 Q_0 + Q_3 Q_0$$

1.0.11 Question 11: Synchronous FSM Sequence Detector '011' [7 Marks]**Problem Statement (2083 Baishakh, Q11)**

Design a synchronous sequential machine that has 1-bit serial input X and output Z which will be HIGH when the input contains the message 011 (Use T Flip-Flop).

Logic Schematic: FSM State Transition Automaton for '011' Sequence Detector

Logic Equations:

$$T_A = BX + A, \quad T_B = \bar{X} + X(A + B), \quad Z = AX$$

Verified Final Result

$$T_A = BX + A, \quad T_B = \bar{X} + X(A + B), \quad Z = AX$$

1.0.12 Question 12: Characteristics of Digital Logic Families [4 Marks]**Problem Statement (2083 Baishakh, Q12)**

Explain about the main characteristics of digital logic families.

1. **Propagation Delay (t_{pd}):** Speed metric of logic switching.
2. **Power Dissipation (P_D):** Power consumed per gate in mW.
3. **Noise Margin (NM):** Maximum tolerable noise voltage.
4. **Fan-Out:** Maximum load gates that can be driven.
5. **Speed-Power Product (SPP):** $SPP = t_{pd} \times P_D$ (pJ).

Chapter 2

2082 Bhadra Examination Solutions

2.0.1 Question 1: Binary to Gray Code Conversion [2 + 2 = 4 Marks]

Problem Statement (2082 Bhadra, Q1)

Convert the following binary codes to Gray codes: (a) 101101_2 , (b) 110001_2 .

Verified Final Result

(a) $(101101)_2 = \mathbf{111011}_{\text{Gray}}$ (b) $(110001)_2 = \mathbf{101001}_{\text{Gray}}$

2.0.2 Question 2: 2's Complement & BCD Addition [1 + 2 = 3 Marks]

Problem Statement (2082 Bhadra, Q2)

What is 2's complement representation? Perform BCD addition of $23 + 48$.

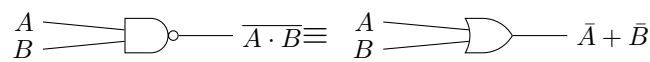
Verified Final Result

$$23 + 48 = \mathbf{0111\ 0001}_{\text{BCD}} = (71)_{10}$$

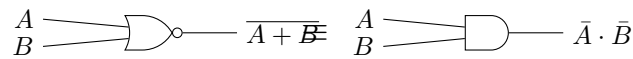
2.0.3 Question 3: De Morgan's Gate Equivalences [4 Marks]

Problem Statement (2082 Bhadra, Q3)

State and explain De Morgan's Theorem with truth table and necessary diagrams.

Logic Schematic: De Morgan's Equivalent Gate Schematics

Theorem 1: $\overline{A \cdot B} = \bar{A} + \bar{B}$



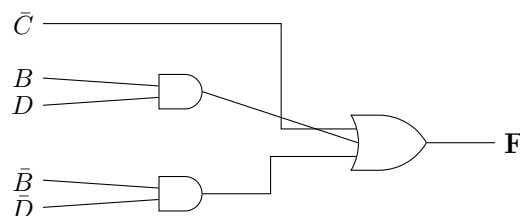
Theorem 2: $\overline{A + B} = \bar{A} \cdot \bar{B}$

2.0.4 Question 4: K-Map Minimization with Don't Cares Circuit [5 Marks]**Problem Statement (2082 Bhadra, Q4)**

Minimize $F(A, B, C, D) = \sum m(0, 1, 2, 4, 7, 8, 9, 10, 12, 15) + d(5, 11, 13)$ using K-Map.

Minimized Boolean Expression:

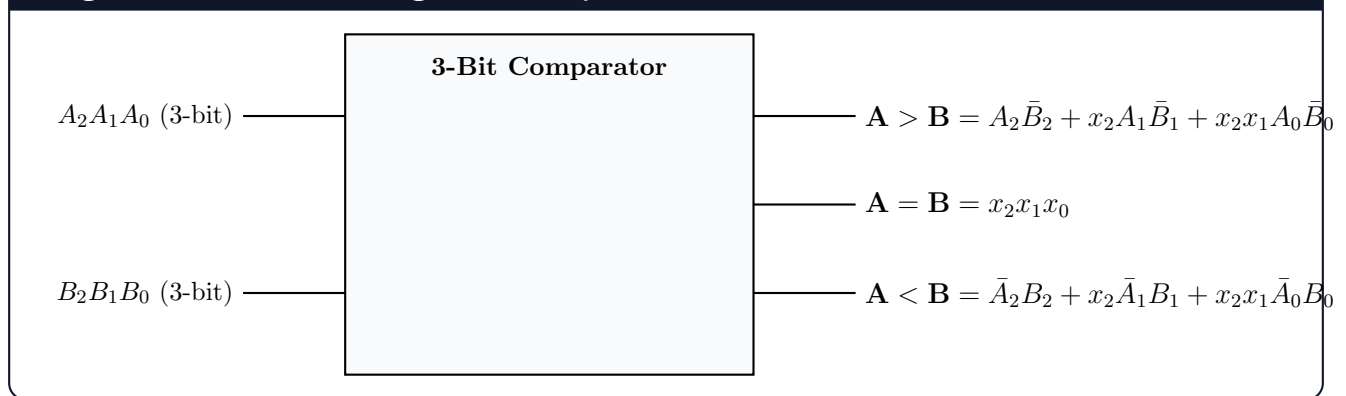
$$F(A, B, C, D) = \bar{C} + BD + \bar{B}\bar{D}$$

Logic Schematic: Logic Realization for $F = \bar{C} + BD + \bar{B}\bar{D}$ **Verified Final Result**

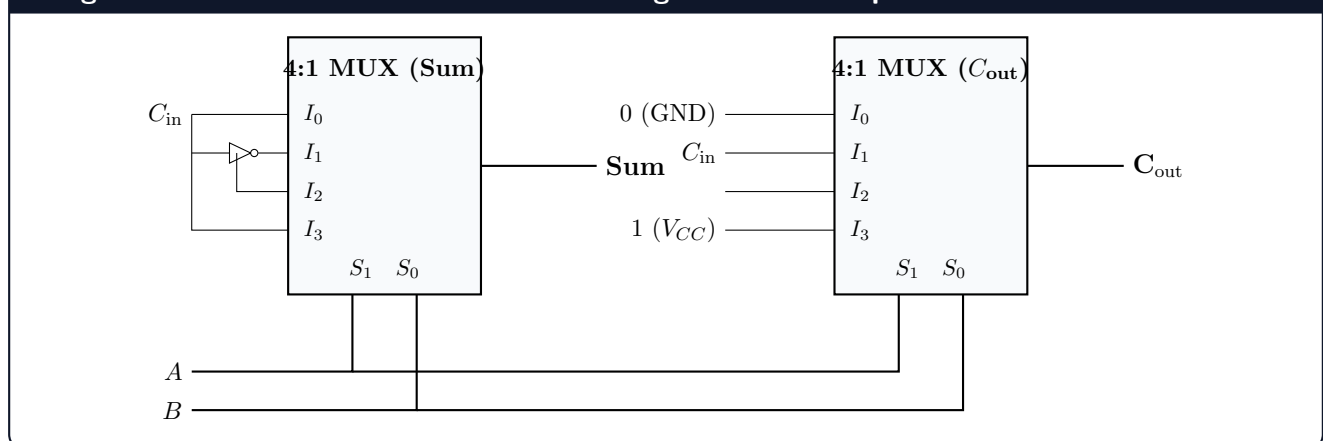
$$\mathbf{F = \bar{C} + BD + \bar{B}\bar{D}}$$

2.0.5 Question 5: 3-Bit Magnitude Comparator Schematic [1 + 5 = 6 Marks]**Problem Statement (2082 Bhadra, Q5)**

What is magnitude comparator? Design a 3-bit magnitude comparator.

Logic Schematic: 3-Bit Magnitude Comparator Functional Schematic**2.0.6 Question 6: Full Adder Using Multiplexers [4 Marks]****Problem Statement (2082 Bhadra, Q6)**

Implement a full adder circuit using Multiplexer.

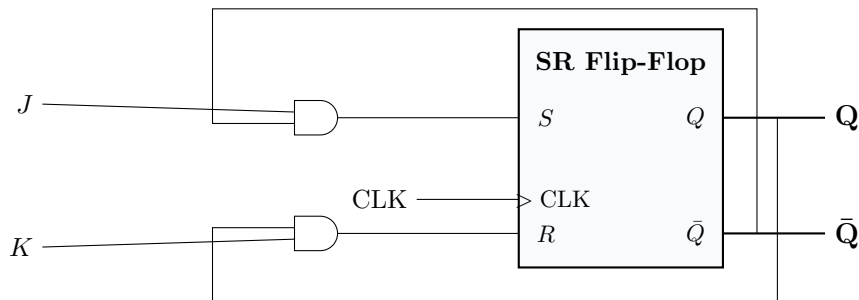
Logic Schematic: Full Adder Realization Using Two 4:1 Multiplexers**Verified Final Result**

Sum: $I_0 = C_{in}, I_1 = \bar{C}_{in}, I_2 = \bar{C}_{in}, I_3 = C_{in}$.

Carry (C_{out}): $I_0 = 0, I_1 = C_{in}, I_2 = C_{in}, I_3 = 1$. Select lines: $S_1 = A, S_0 = B$.

2.0.7 Question 7: SR to JK Flip-Flop Conversion Schematic [2 + 5 = 7 Marks]**Problem Statement (2082 Bhadra, Q7)**

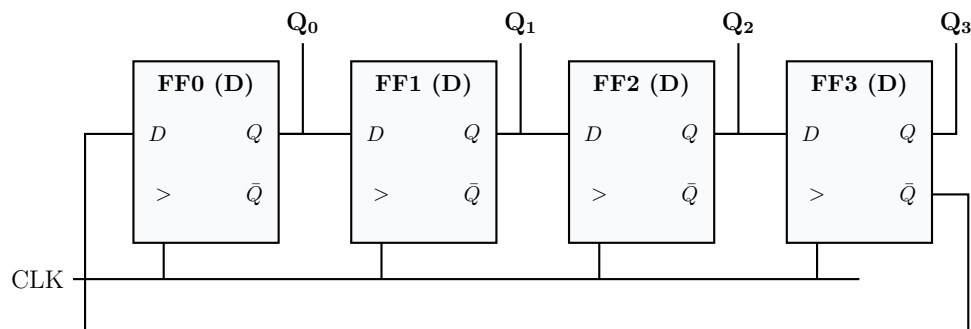
Differentiate between combinational and sequential circuits. Convert SR flip-flop into JK flip-flop.

Logic Schematic: JK Flip-Flop Implemented Using SR Flip-Flop**Verified Final Result**

$$S = J\bar{Q}, \quad R = KQ$$

2.0.8 Question 8: 4-Bit Johnson Counter Schematic [2 + 3 = 5 Marks]**Problem Statement (2082 Bhadra, Q8)**

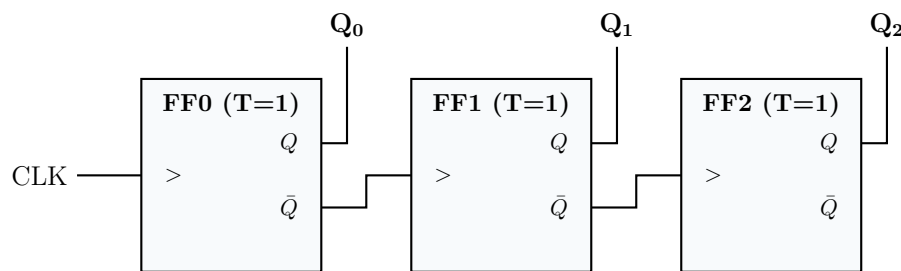
Write briefly about types of shift registers. Explain the operation of a Johnson's Counter.

Logic Schematic: 4-Bit Johnson Counter (Twisted Ring Counter) Schematic**Verified Final Result**

$2n = 8$ states: 0000 $\rightarrow$ 1000 $\rightarrow$ 1100 $\rightarrow$ 1110 $\rightarrow$ 1111 $\rightarrow$ 0111 $\rightarrow$ 0011 $\rightarrow$ 0001 $\rightarrow$ 0000.

2.0.9 Question 9: 3-Bit Ripple UP Counter with Positive Edge Triggering [5 Marks]**Problem Statement (2082 Bhadra, Q9)**

Describe the operation of 3-bit ripple up counter with positive edge triggered clock.

Logic Schematic: 3-Bit Ripple UP Counter Schematic (Positive Edge Triggered)**Verified Final Result**

Counts $000 \rightarrow 111$. Each stage toggles on the positive clock edge produced by preceding $\bar{Q}$.

2.0.10 Question 10: Synchronous Mod-6 Counter Using SR Flip-Flops [7 Marks]**Problem Statement (2082 Bhadra, Q10)**

Design the synchronous mod-6 counter using SR flip-flop and draw its timing diagram.

Design Equations:

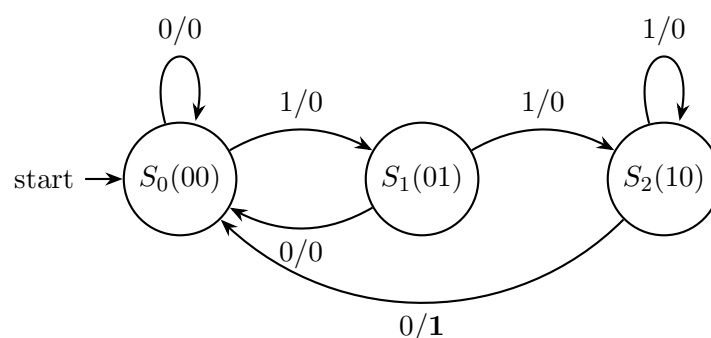
$$S_0 = \bar{Q}_0, \quad R_0 = Q_0, \quad S_1 = \bar{Q}_2 \bar{Q}_1 Q_0, \quad R_1 = Q_1 Q_0, \quad S_2 = Q_1 Q_0, \quad R_2 = Q_0$$

Verified Final Result

$$S_0 = \bar{Q}_0, \quad R_0 = Q_0, \quad S_1 = \bar{Q}_2 \bar{Q}_1 Q_0, \quad R_1 = Q_1 Q_0, \quad S_2 = Q_1 Q_0, \quad R_2 = Q_0$$

2.0.11 Question 11: Synchronous FSM Sequence Detector '110' [7 Marks]**Problem Statement (2082 Bhadra, Q11)**

Design a synchronous sequential machine that has 1-bit serial input X and output Z which will be HIGH when the input contains the message 110 (Use SR Flip-Flop).

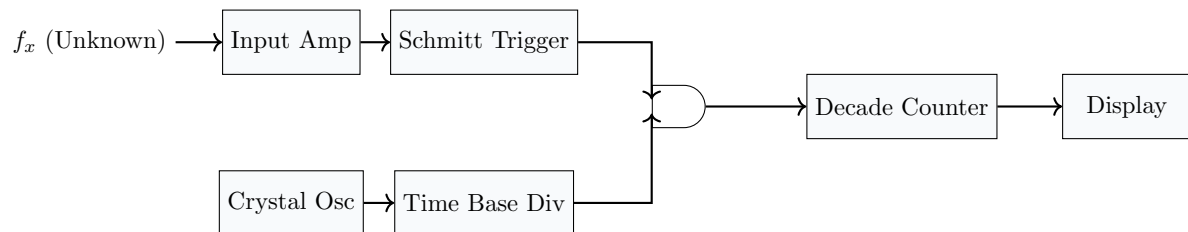
Logic Schematic: FSM State Transition Automaton for '110' Sequence Detector

Verified Final Result

$$S_A = BX, \quad R_A = \bar{X}, \quad S_B = \bar{A}\bar{B}X, \quad R_B = B\bar{X} + A, \quad Z = A\bar{X}$$

2.0.12 Question 12: Frequency Counter Block Diagram [3 Marks]**Problem Statement (2082 Bhadra, Q12)**

Write a short note on frequency counter with functional block diagram.

Logic Schematic: Digital Frequency Counter Functional Block Diagram

Chapter 3

2082 Baishakh Examination Solutions

3.0.1 Question 1: ASCII Code & 2's Complement Addition [1 + 3 = 4 Marks]

Problem Statement (2082 Baishakh, Q1)

Define an ASCII code. Use 2's complement method to perform $(-28 + 12)_{10}$ in 8-bit signed number representation.

Verified Final Result

$$(-28 + 12)_{10} = 1111\ 0000_2 = -16_{10}$$

3.0.2 Question 2: Boolean Algebra Proofs [2 + 2 = 4 Marks]

Problem Statement (2082 Baishakh, Q2)

Prove: (a) $AB + \bar{A}C + BC = AB + \bar{A}C$, (b) $AB + C(A \oplus B) = BC + A(B + C)$.

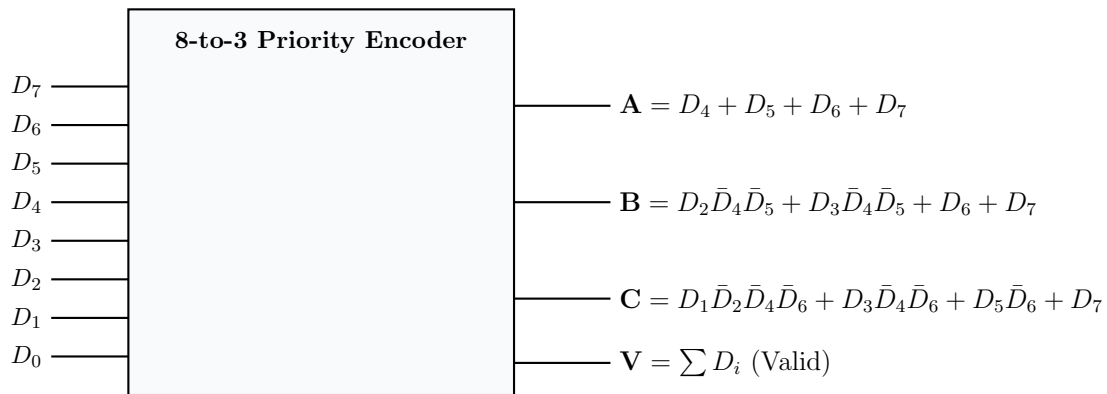
Verified Final Result

Both identities are verified algebraically.

3.0.3 Question 3: Octal Priority Encoder Schematic [2 + 6 = 8 Marks]

Problem Statement (2082 Baishakh, Q3)

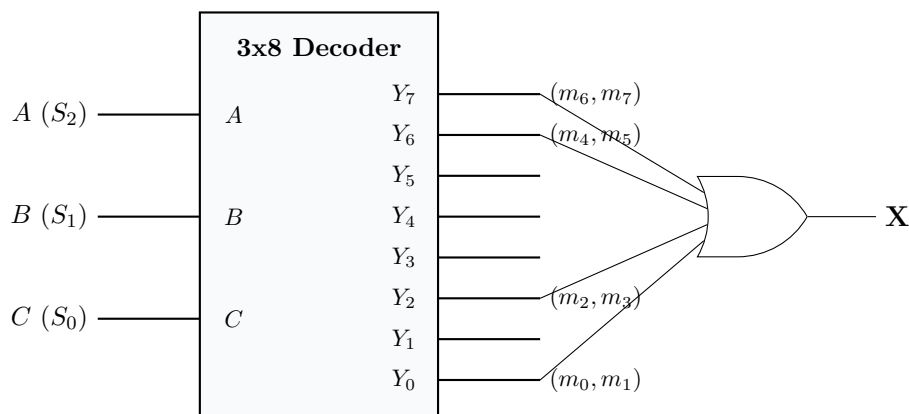
What is a decoder? Design an octal priority encoder with neat circuit diagram.

Logic Schematic: 8-to-3 Octal Priority Encoder Logic Schematic**3.0.4 Question 4: 3x8 Decoder Function Realization [3 + 3 = 6 Marks]****Problem Statement (2082 Baishakh, Q4)**

Realize $X(A, B, C, D) = \sum m(0, 2, 3, 7, 8, 10, 11, 14, 15)$ using a single 3x8 decoder and simplify implementing K-Map.

K-Map Simplification:

$$X(A, B, C, D) = \bar{B}\bar{D} + \bar{B}C + CD + ABC$$

Logic Schematic: Realization of 4-Variable Function Using 3x8 Decoder**Verified Final Result**

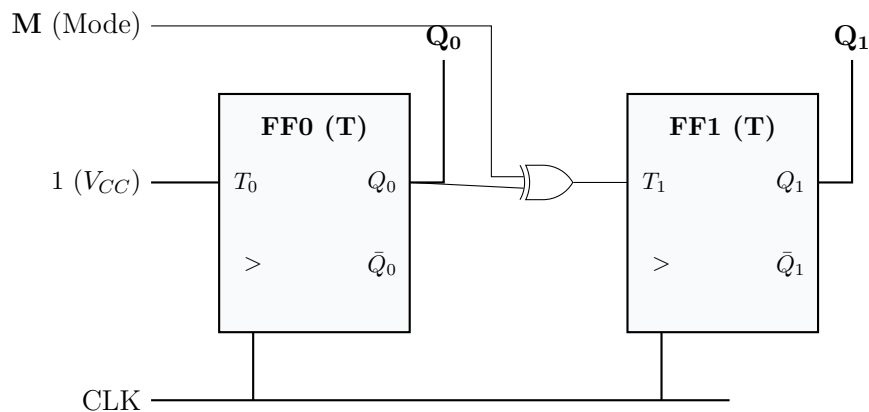
$$X(A, B, C, D) = \bar{B}\bar{D} + \bar{B}C + CD + ABC$$

3.0.5 Question 5: Synchronous 2-Bit UP/DOWN Counter Schematic [7 Marks]**Problem Statement (2082 Baishakh, Q5)**

Design a synchronous 2-bit up/down counter using T flip-flops.

Design Equations:

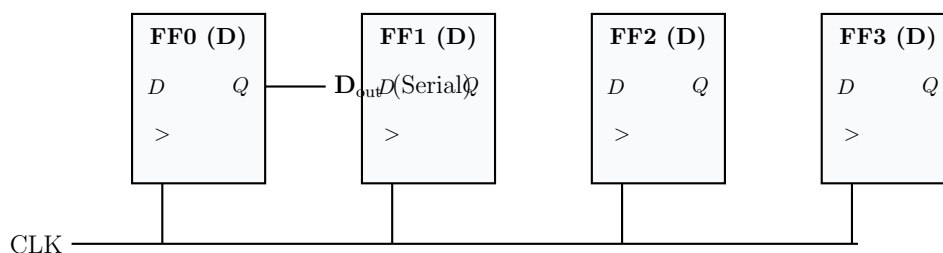
$$T_0 = 1, \quad T_1 = M \oplus Q_0$$

Logic Schematic: Synchronous 2-Bit UP/DOWN Counter Schematic**Verified Final Result**

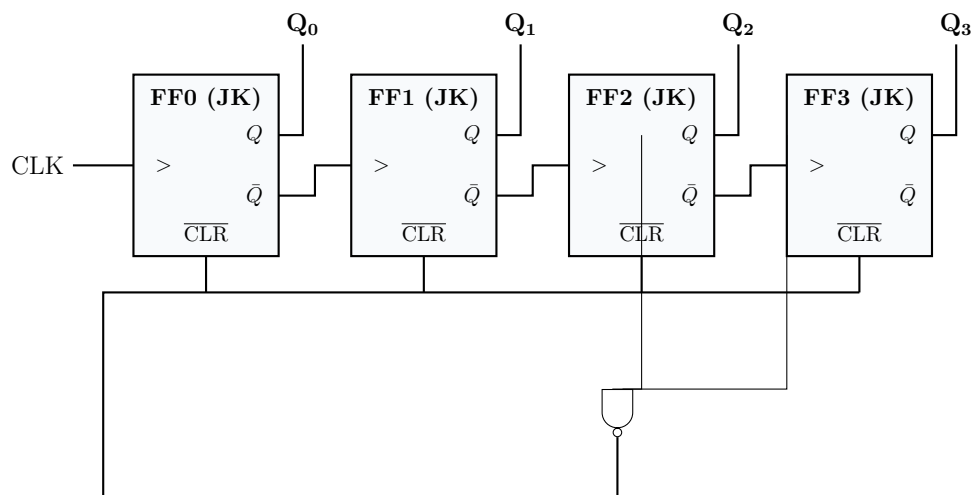
$$T_0 = 1, \quad T_1 = M \oplus Q_0$$

3.0.6 Question 6: 4-Bit PISO Shift Register Schematic [3 + 2 = 5 Marks]**Problem Statement (2082 Baishakh, Q6)**

Explain the operation of 4-bit parallel-in serial-out (PISO) shift register with necessary circuit and timing diagram for 1101 input data.

Logic Schematic: 4-Bit Parallel-In Serial-Out (PISO) Shift Register Logic Diagram**3.0.7 Question 7: Mod-12 Asynchronous Counter Schematic [3 + 3 = 6 Marks]****Problem Statement (2082 Baishakh, Q7)**

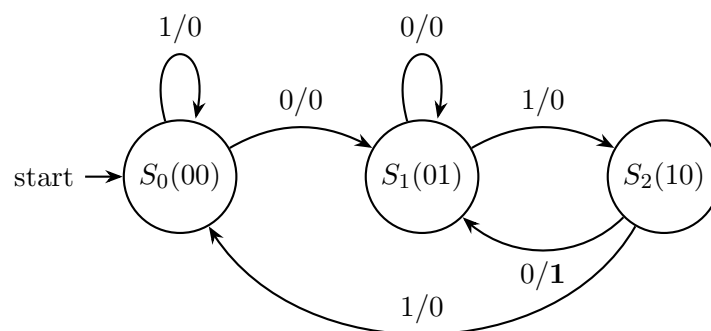
Sketch the circuit diagram of mod-12 asynchronous counter having positive-edge triggering clock system implementing JK flip-flops with neat timing diagram.

Logic Schematic: Mod-12 Asynchronous Counter (Positive Edge Triggered)**Verified Final Result**

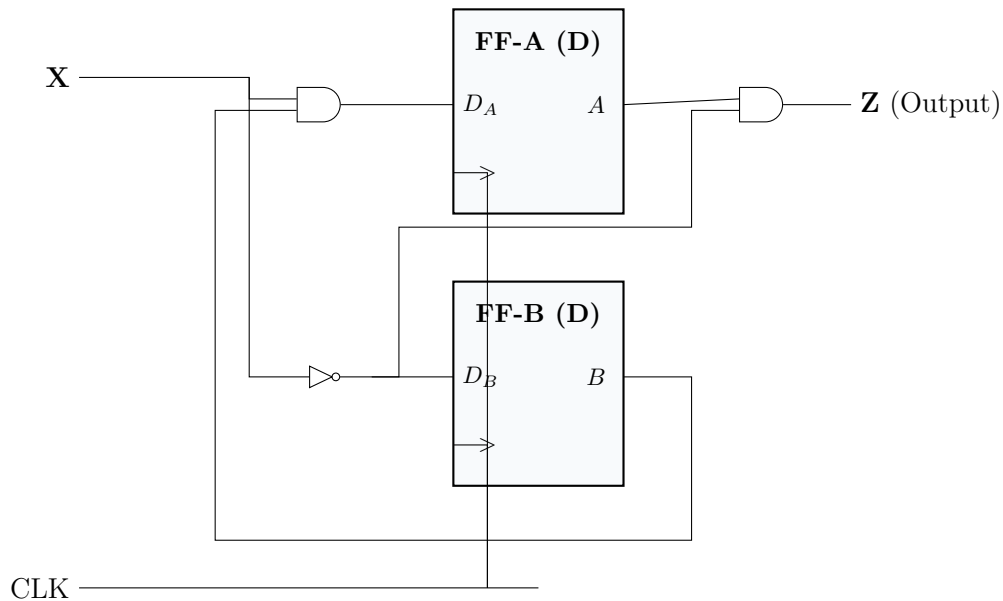
Reset Logic: $\overline{\text{CLR}} = \overline{Q_3 \cdot Q_2}$. Modulus = 12 (0000 to 1011).

3.0.8 Question 8: FSM Sequence Detector '010' Using D Flip-Flops [10 Marks]**Problem Statement (2082 Baishakh, Q8)**

Design a sequential machine that consists of one input X and one output Z . $Z = 1$ whenever it detects the serial sequence 010. Implement only D flip-flops.

Logic Schematic: FSM State Transition Automaton for '010' Sequence Detector**Design Equations:**

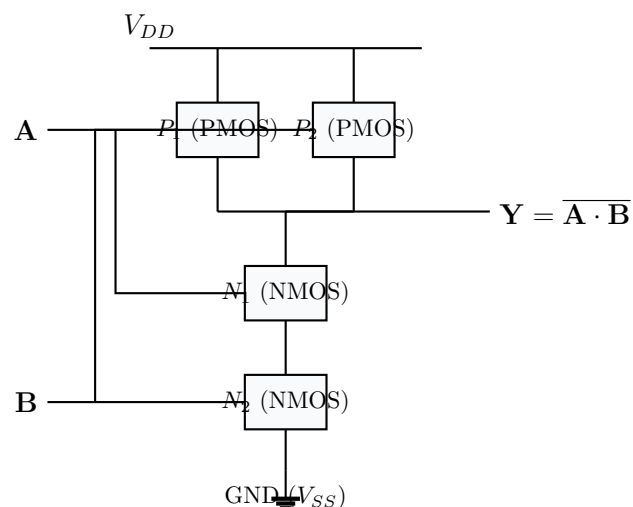
$$D_A = BX, \quad D_B = \bar{X}, \quad Z = A\bar{X}$$

Logic Schematic: Logic Schematic with D Flip-Flops for '010' Detector**Verified Final Result**

$$D_A = BX, \quad D_B = \bar{X}, \quad Z = A\bar{X}$$

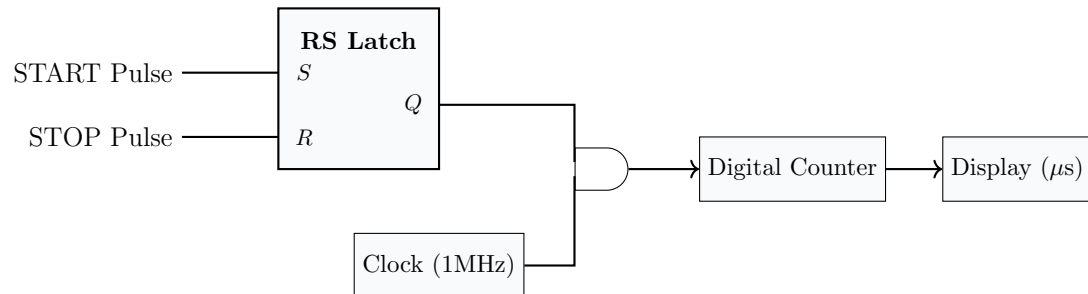
3.0.9 Question 9: Two-Input CMOS NAND Gate Transistor Circuit [4 + 2 = 6 Marks]**Problem Statement (2082 Baishakh, Q9)**

Draw the circuit diagram of two-input CMOS NAND gate, explain its logic operation, and list the characteristics of TTL logic family.

Logic Schematic: Two-Input CMOS NAND Gate Transistor Schematic

3.0.10 Question 10: Time Interval Measurement Block Diagram [4 Marks]**Problem Statement (2082 Baishakh, Q10)**

With the help of functional diagram explain the operation of time measuring circuit.

Logic Schematic: Time Interval Measurement System Block Diagram

Chapter 4

2081 Ashwin Examination Solutions

4.0.1 Question 1: Merits & Demerits of Digital Signals [2 Marks]

Problem Statement (2081 Ashwin, Q1)

Mention merits and demerits of digital signal over analog signal.

Merits: High noise immunity, easy error detection/correction, perfect storage/retrieval, compact digital ICs.

Demerits: Higher transmission bandwidth, quantization noise in ADC.

4.0.2 Question 2: Radix Gray Code Conversions [3 Marks]

Problem Statement (2081 Ashwin, Q2)

Perform: (a) $(6E.2C)_{16} = (?)_8$, (b) $(10110110)_{\text{Gray}} = (?)_2$.

Verified Final Result

(a) $(6E.2C)_{16} = (156.13)_8$ (b) $(10110110)_{\text{Gray}} = (11011011)_2$

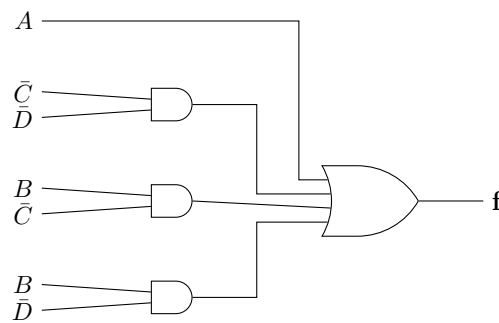
4.0.3 Question 3: BCD to 7-Segment 'f' Segment Decoder Logic Circuit [4 Marks]

Problem Statement (2081 Ashwin, Q3)

Design the simplest logic circuit for 'f' segment for the BCD-to-seven segment display decoder.

Minimized Boolean Equation:

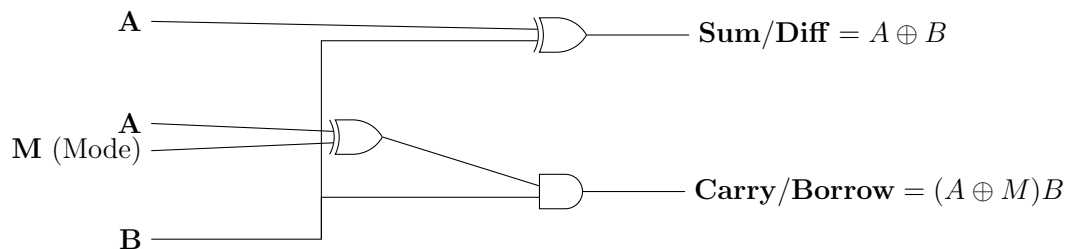
$$f = A + \bar{C}\bar{D} + B\bar{C} + B\bar{D}$$

Logic Schematic: Logic Gate Circuit for 7-Segment 'f' Segment**Verified Final Result**

$$f = A + \bar{C}\bar{D} + B\bar{C} + B\bar{D}$$

4.0.4 Question 4: 4:1 MUX Function Realization Combined Adder/Subtractor [3 + 7 = 10 Marks]**Problem Statement (2081 Ashwin, Q4)**

Implement $Y(A, B, C) = \sum m(0, 2, 3, 5, 7)$ using only a single 4:1 MUX. Design a combined Half-Adder and Half-Subtractor with a mode control.

Logic Schematic: Combined Half-Adder and Half-Subtractor Logic Circuit**Verified Final Result**

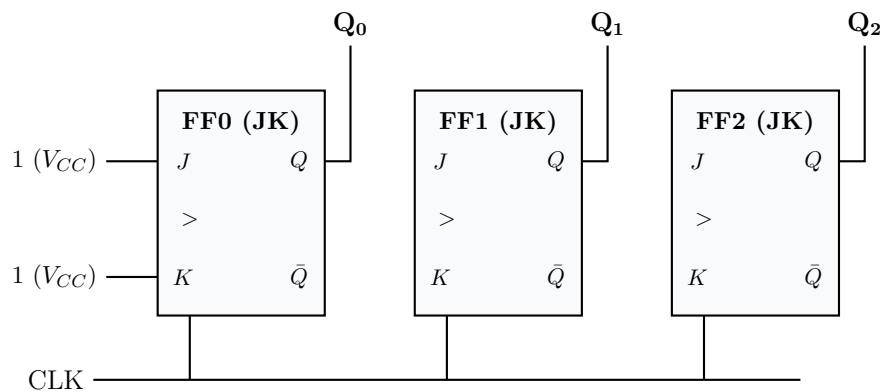
$M = 0 \implies$ Half Adder; $M = 1 \implies$ Half Subtractor.

4.0.5 Question 5: Mod-6 Synchronous DOWN Counter Schematic [7 Marks]**Problem Statement (2081 Ashwin, Q5)**

Design a mod-6 synchronous down counter using JK flip-flops ($5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0 \rightarrow 5$).

Design Equations:

$$J_0 = 1, K_0 = 1, \quad J_1 = \bar{Q}_2\bar{Q}_0, K_1 = \bar{Q}_0, \quad J_2 = K_2 = \bar{Q}_1\bar{Q}_0$$

Logic Schematic: Mod-6 Synchronous DOWN Counter Schematic**Verified Final Result**

$$J_0 = 1, K_0 = 1, J_1 = \bar{Q}_2 \bar{Q}_0, K_1 = \bar{Q}_0, J_2 = K_2 = \bar{Q}_1 \bar{Q}_0$$

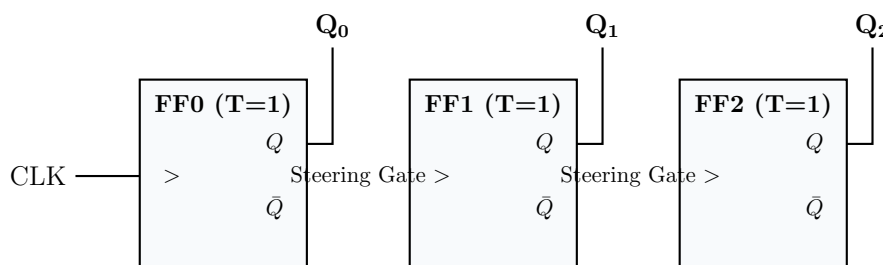
4.0.6 Question 6: 4-Bit SIPO Shift Register Schematic [3 + 3 = 6 Marks]**Problem Statement (2081 Ashwin, Q6)**

Explain the operation of 4-bit serial-in parallel-out (SIPO) shift register with necessary circuit and timing diagram for input data of 1011.

Four clock cycles load the serial sequence 1011_2 into parallel outputs $Q_3Q_2Q_1Q_0 = 1101_2$.

4.0.7 Question 7: 3-Bit Asynchronous UP/DOWN Counter Schematic [4 + 3 = 7 Marks]**Problem Statement (2081 Ashwin, Q7)**

Describe briefly the operation of 3 bit up/down asynchronous counter having negative edge triggering clock system with neat circuit diagram and timing diagram.

Logic Schematic: 3-Bit Asynchronous UP/DOWN Counter with Negative Edge Triggering**Verified Final Result**

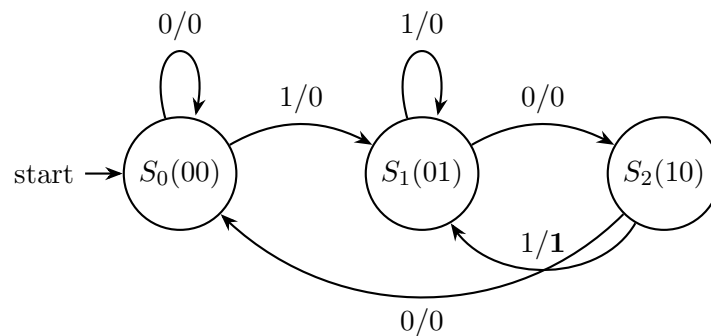
Steering Logic: $CLK_{i+1} = \bar{M}Q_i + M\bar{Q}_i$. $M = 0 \Rightarrow$ UP Counter, $M = 1 \Rightarrow$ DOWN Counter.

4.0.8 Question 8: FSM Sequence Detector '101' Using SR Flip-Flops [10 Marks]

Problem Statement (2081 Ashwin, Q8)

Design a sequential machine that consists of one input X and one output Z . $Z = 1$ whenever it detects the serial sequence 101. Implement only SR flip-flops.

Logic Schematic: FSM State Transition Automaton for '101' Sequence Detector



Design Equations:

$$S_A = BX, \quad R_A = \bar{X}, \quad S_B = X(\bar{A}\bar{B} + A), \quad R_B = B\bar{X}, \quad Z = AX$$

Verified Final Result

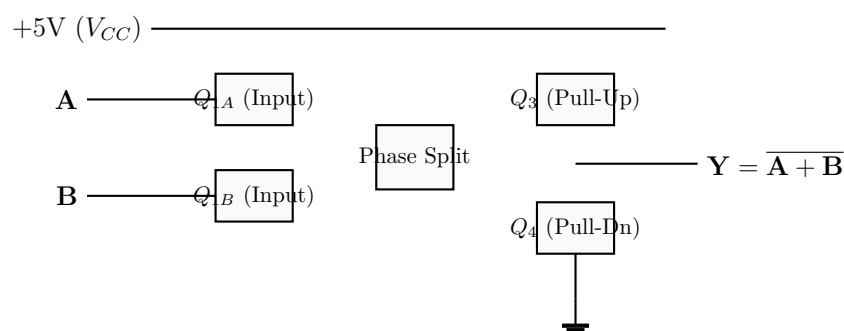
$$S_A = BX, \quad R_A = \bar{X}, \quad S_B = X(\bar{A}\bar{B} + A), \quad R_B = B\bar{X}, \quad Z = AX$$

4.0.9 Question 9: Two-Input TTL NOR Gate Transistor Circuit [5 + 2 = 7 Marks]

Problem Statement (2081 Ashwin, Q9)

Draw the circuit diagram of two-input TTL NOR gate, explain its operation, and list the characteristics of CMOS logic family.

Logic Schematic: Two-Input TTL NOR Gate Schematic (Totem-Pole Output)

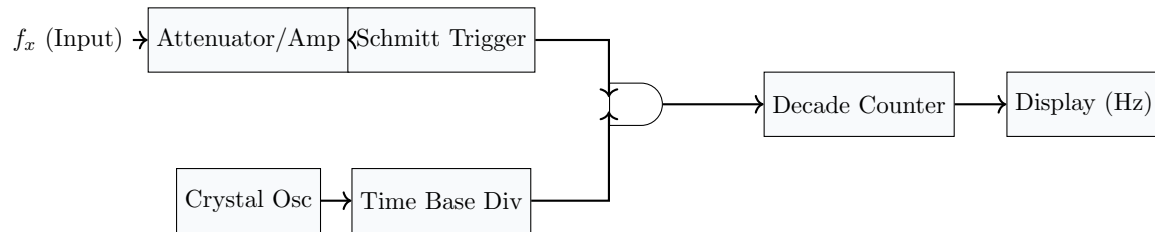


4.0.10 Question 10: Digital Frequency Measurement Block Diagram [4 Marks]

Problem Statement (2081 Ashwin, Q10)

With the help of functional diagram explain the operation of frequency measurement.

Logic Schematic: Digital Frequency Measurement System Block Diagram



Verified Final Result

$$f_x = \frac{N}{T_{\text{gate}}}$$